

Merge Sort: Runtime Analysis for Beginners

What is Merge Sort?

Merge Sort is a **divide and conquer** algorithm. It works by:

1. Dividing the array into halves until each subarray has one element.
2. Merging the subarrays back together in sorted order.

Runtime Analysis

Step 1: Number of Splits

Each time we split the array in half. We continue this until each part has only one element. This takes:

$$\log_2 n \text{ splits (levels of recursion)}$$

Step 2: Work at Each Level

At each level, all elements are merged. Merging takes linear time:

$$O(n) \text{ work per level}$$

Total Runtime

There are $\log n$ levels, and each level does $O(n)$ work:

$$\text{Total time} = O(n \log n)$$

Summary

Step	Work Done
Divide	$\log n$ levels
Merge	$O(n)$ per level
Total Runtime	$O(n \log n)$

Why is Merge Sort Efficient?

- Faster than Bubble Sort and Insertion Sort ($O(n^2)$).
- Works well for large datasets.
- Stable sorting algorithm.
- Needs extra space for merging.